Pandas

August 27, 2023

1 Pandas

1.1 Why Pandas ?

- Easy to Use
- Intergrated with other ML frameworks
- Helps getting Data Ready for ML

1.2 Sections :

- Most used functions
- Pandas DataTypes
- Importing and Exporting Data
- Describing data
- Viewing and Selecting Data
- Manipulating Data

1.3 Links :

- PandasDoc

[1]:
```
import pandas as pd
```

2 Series

[2]:
```
# Series -> 1D Data
series = pd.Series(["BMW", "Toyota", "Honda"])
```

[3]:
```
series
```

[3]:
```
0       BMW
1    Toyota
2     Honda
dtype: object
```

3 DataFrame

[4]:
```
# DataFrame -> 2D Data
color =  pd.Series(["Black", "Blue", "Green"])
car_data = pd.DataFrame({"Car Make": series, "Color": color})
car_data.head()
```

[4]:
```
  Car Make  Color
0      BMW  Black
1   Toyota   Blue
2    Honda  Green
```

4 Import Data

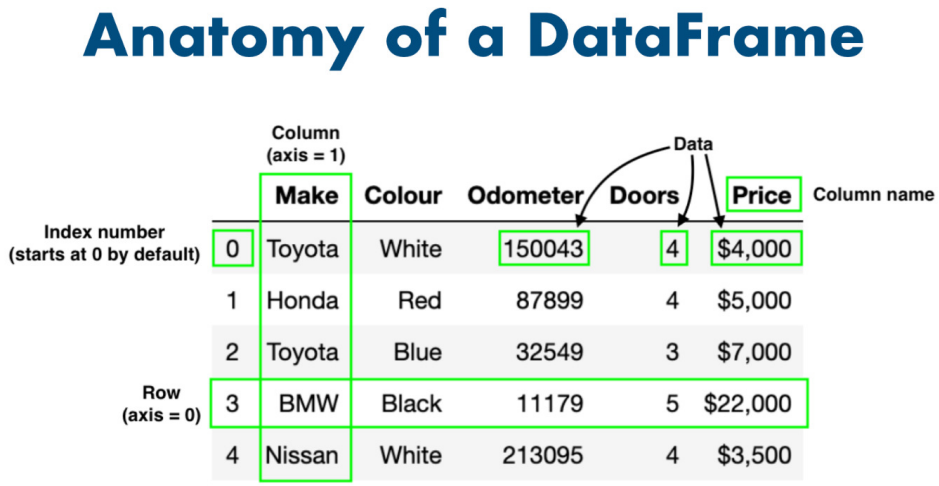
[5]:
```
# read csv data
df = pd.read_csv("resources/car-sales.csv")
```

[6]:
```
df
```

[6]:

	Make	Colour	Odometer (KM)	Doors	Price
0	Toyota	White	150043	4	$4,000.00
1	Honda	Red	87899	4	$5,000.00
2	Toyota	Blue	32549	3	$7,000.00
3	BMW	Black	11179	5	$22,000.00
4	Nissan	White	213095	4	$3,500.00
5	Toyota	Green	99213	4	$4,500.00
6	Honda	Blue	45698	4	$7,500.00
7	Honda	Blue	54738	4	$7,000.00
8	Toyota	White	60000	4	$6,250.00
9	Nissan	White	31600	4	$9,700.00

5 Understanding DataFrame



[7]:
```
# exporting dataframe
df.to_csv("test.csv", index=False)
```

[8]:
```
new_df = pd.read_csv("test.csv")
new_df
```

[8]:

	Make	Colour	Odometer (KM)	Doors	Price
0	Toyota	White	150043	4	$4,000.00
1	Honda	Red	87899	4	$5,000.00
2	Toyota	Blue	32549	3	$7,000.00
3	BMW	Black	11179	5	$22,000.00
4	Nissan	White	213095	4	$3,500.00
5	Toyota	Green	99213	4	$4,500.00
6	Honda	Blue	45698	4	$7,500.00
7	Honda	Blue	54738	4	$7,000.00
8	Toyota	White	60000	4	$6,250.00
9	Nissan	White	31600	4	$9,700.00

6 Describe Data

[9]:
```
# attributes
new_df.dtypes
```

[9]:
```
Make              object
Colour            object
```

```
Odometer (KM)     int64
Doors             int64
Price            object
dtype: object
```

[10]:
```
# Function
new_df.describe()
```

[10]:
```
       Odometer (KM)      Doors
count      10.000000  10.000000
mean    78601.400000   4.000000
std     61983.471735   0.471405
min     11179.000000   3.000000
25%     35836.250000   4.000000
50%     57369.000000   4.000000
75%     96384.500000   4.000000
max    213095.000000   5.000000
```

[11]:
```
new_df.index
```

[11]:
```
RangeIndex(start=0, stop=10, step=1)
```

[12]:
```
new_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   Make           10 non-null     object
 1   Colour         10 non-null     object
 2   Odometer (KM)  10 non-null     int64 
 3   Doors          10 non-null     int64 
 4   Price          10 non-null     object
dtypes: int64(2), object(3)
memory usage: 528.0+ bytes
```

[13]:
```
# Selecting single Column
new_df['Doors'].mean()
```

[13]:
```
4.0
```

[14]:
```
# Size of DataFrame
new_df.shape
```

[14]:
```
(10, 5)
```

6.1 Viewing and Selecting Data

[15]:
```
# Returns just top 5 rows in DataFrame
new_df.head()
```

[15]:

	Make	Colour	Odometer (KM)	Doors	Price
0	Toyota	White	150043	4	$4,000.00
1	Honda	Red	87899	4	$5,000.00
2	Toyota	Blue	32549	3	$7,000.00
3	BMW	Black	11179	5	$22,000.00
4	Nissan	White	213095	4	$3,500.00

[16]:
```
new_df.head(7)
```

[16]:

	Make	Colour	Odometer (KM)	Doors	Price
0	Toyota	White	150043	4	$4,000.00
1	Honda	Red	87899	4	$5,000.00
2	Toyota	Blue	32549	3	$7,000.00
3	BMW	Black	11179	5	$22,000.00
4	Nissan	White	213095	4	$3,500.00
5	Toyota	Green	99213	4	$4,500.00
6	Honda	Blue	45698	4	$7,500.00

[17]:
```
# Return just bottom 5 rows in DataFrame
new_df.tail()
```

[17]:

	Make	Colour	Odometer (KM)	Doors	Price
5	Toyota	Green	99213	4	$4,500.00
6	Honda	Blue	45698	4	$7,500.00
7	Honda	Blue	54738	4	$7,000.00
8	Toyota	White	60000	4	$6,250.00
9	Nissan	White	31600	4	$9,700.00

[18]:
```
new_df.tail(7)
```

[18]:

	Make	Colour	Odometer (KM)	Doors	Price
3	BMW	Black	11179	5	$22,000.00
4	Nissan	White	213095	4	$3,500.00
5	Toyota	Green	99213	4	$4,500.00
6	Honda	Blue	45698	4	$7,500.00
7	Honda	Blue	54738	4	$7,000.00
8	Toyota	White	60000	4	$6,250.00
9	Nissan	White	31600	4	$9,700.00

7 Loc and Iloc

[19]:
```
# Get element by index -> loc
new_df.loc[6]
```

[19]:
```
Make                 Honda
Colour                Blue
Odometer (KM)        45698
Doors                    4
Price            $7,500.00
Name: 6, dtype: object
```

[20]:
```
# Get element by position -> iloc
new_df.iloc[6]
```

[20]:
```
Make                 Honda
Colour                Blue
Odometer (KM)        45698
Doors                    4
Price            $7,500.00
Name: 6, dtype: object
```

[21]:
```
colors = pd.Series(["red","blue","green","black","white","yellow",],␣
↪index=[1,3,5,4,6,8])
```

[22]:
```
colors
```

[22]:
```
1       red
3      blue
5     green
4     black
6     white
8    yellow
dtype: object
```

[23]:
```
colors.loc[3]
```

[23]:
```
'blue'
```

[24]:
```
colors.iloc[3]
```

[24]:
```
'black'
```

[25]:
```
# upto 3
colors.iloc[:3]
```

[25]:
```
1      red
3     blue
5    green
dtype: object
```

[26]:
```
# upto and including 3
colors.loc[:3]
```

[26]:
```
1     red
3    blue
dtype: object
```

[27]:
```
# Two ways to Select Column (Note :: Won't work when space is present in column name)
new_df.Make
```

[27]:
```
0    Toyota
1     Honda
2    Toyota
3       BMW
4    Nissan
5    Toyota
6     Honda
7     Honda
8    Toyota
9    Nissan
Name: Make, dtype: object
```

[28]:
```
new_df['Make']
```

[28]:
```
0    Toyota
1     Honda
2    Toyota
3       BMW
4    Nissan
5    Toyota
6     Honda
7     Honda
8    Toyota
9    Nissan
Name: Make, dtype: object
```

[29]:
```
# Filter Columns
new_df[new_df['Make'] == 'Toyota']
```

[29]:

	Make	Colour	Odometer (KM)	Doors	Price
0	Toyota	White	150043	4	$4,000.00

```
2  Toyota   Blue          32549      3  $7,000.00
5  Toyota  Green          99213      4  $4,500.00
8  Toyota  White          60000      4  $6,250.00
```

[30]:
```
new_df[new_df['Odometer (KM)'] >= 100000]
```

[30]:
```
     Make Colour  Odometer (KM)  Doors      Price
0  Toyota  White         150043      4  $4,000.00
4  Nissan  White         213095      4  $3,500.00
```

[31]:
```
new_df
```

[31]:
```
     Make Colour  Odometer (KM)  Doors       Price
0  Toyota  White         150043      4   $4,000.00
1   Honda    Red          87899      4   $5,000.00
2  Toyota   Blue          32549      3   $7,000.00
3     BMW  Black          11179      5  $22,000.00
4  Nissan  White         213095      4   $3,500.00
5  Toyota  Green          99213      4   $4,500.00
6   Honda   Blue          45698      4   $7,500.00
7   Honda   Blue          54738      4   $7,000.00
8  Toyota  White          60000      4   $6,250.00
9  Nissan  White          31600      4   $9,700.00
```

[32]:
```
pd.crosstab(new_df['Make'], new_df['Doors'])
```

[32]:
```
Doors   3  4  5
Make
BMW     0  0  1
Honda   0  3  0
Nissan  0  2  0
Toyota  1  3  0
```

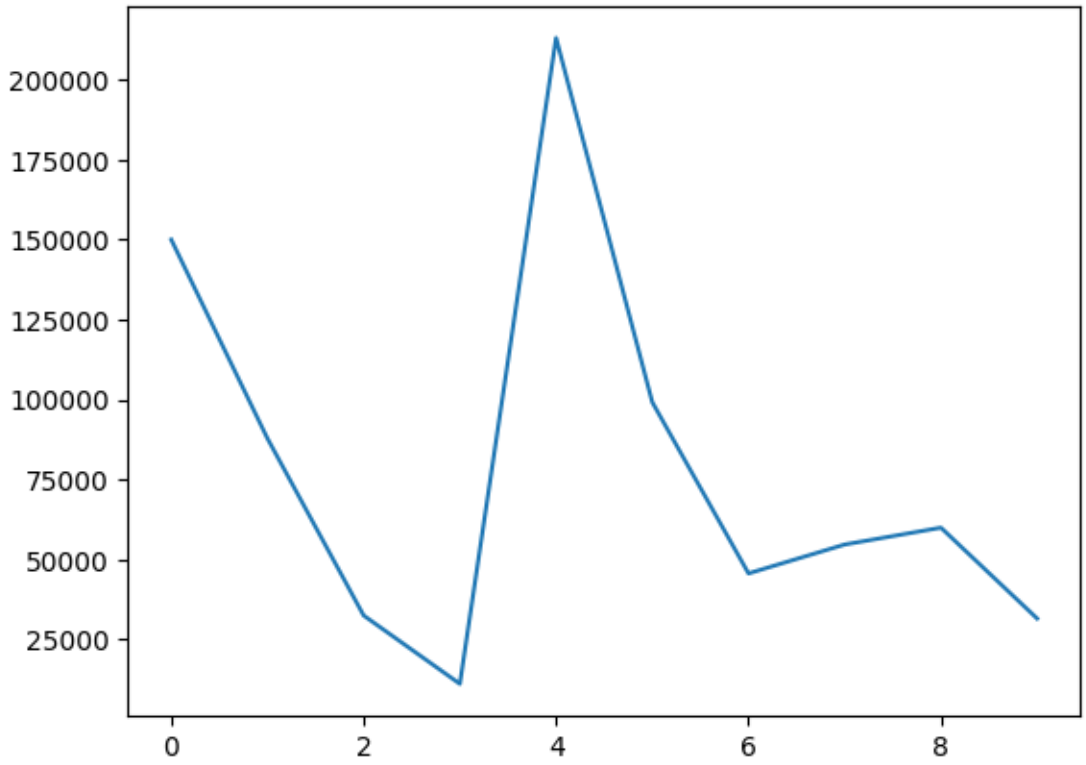
[33]:
```
new_df.groupby(by="Make").mean(numeric_only=True)
```

[33]:
```
        Odometer (KM)  Doors
Make
BMW       11179.000000   5.00
Honda     62778.333333   4.00
Nissan   122347.500000   4.00
Toyota    85451.250000   3.75
```

[34]:
```
# Visualize
"""
%matplotlib inline
import matplotlib.pyplot as plt
"""
```

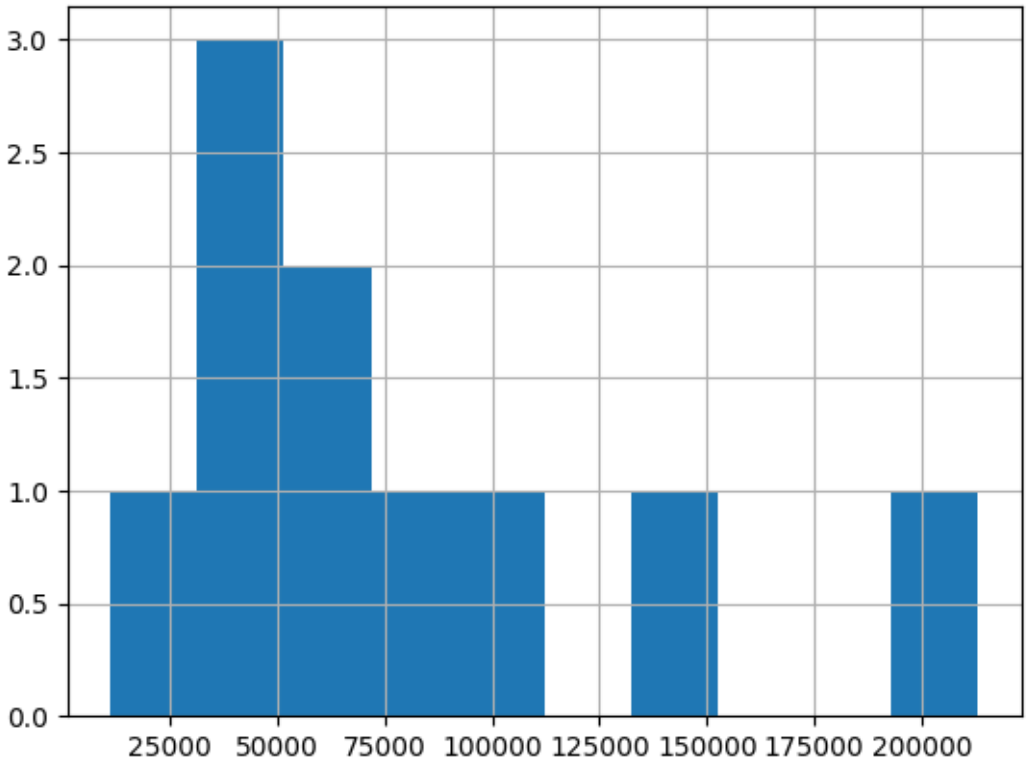
```
new_df["Odometer (KM)"].plot()
```

[34]: <Axes: >



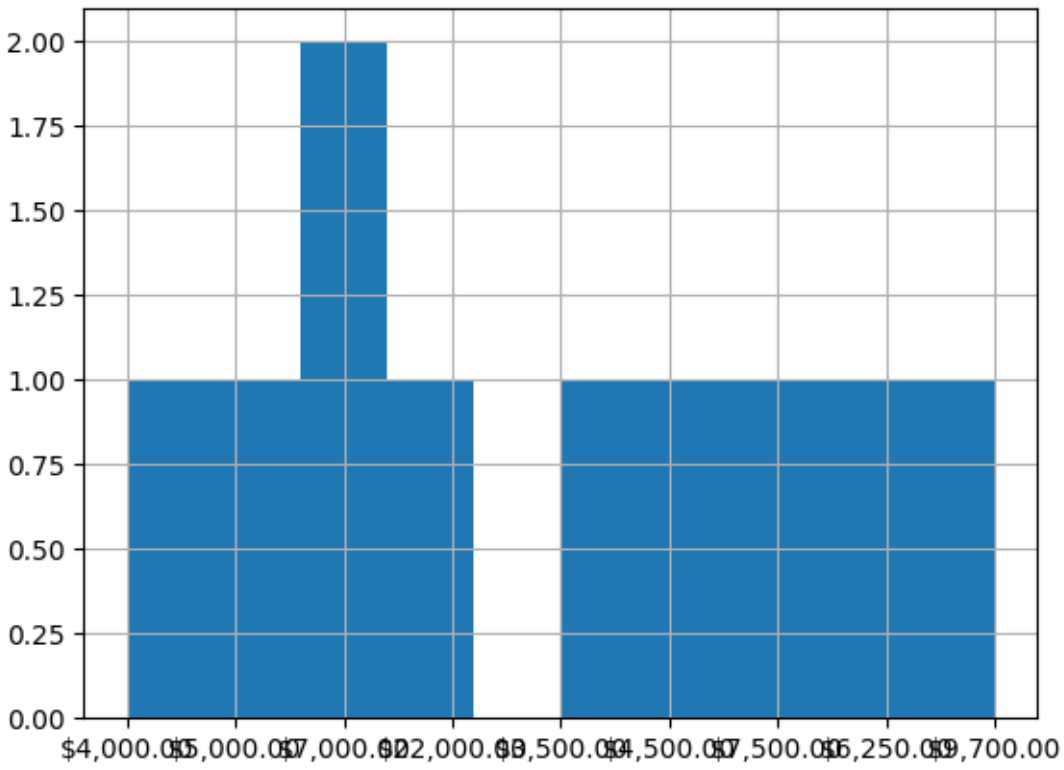
[35]:
```
new_df["Odometer (KM)"].hist()
```

[35]: <Axes: >



[36]:
```
new_df["Price"].hist()
```

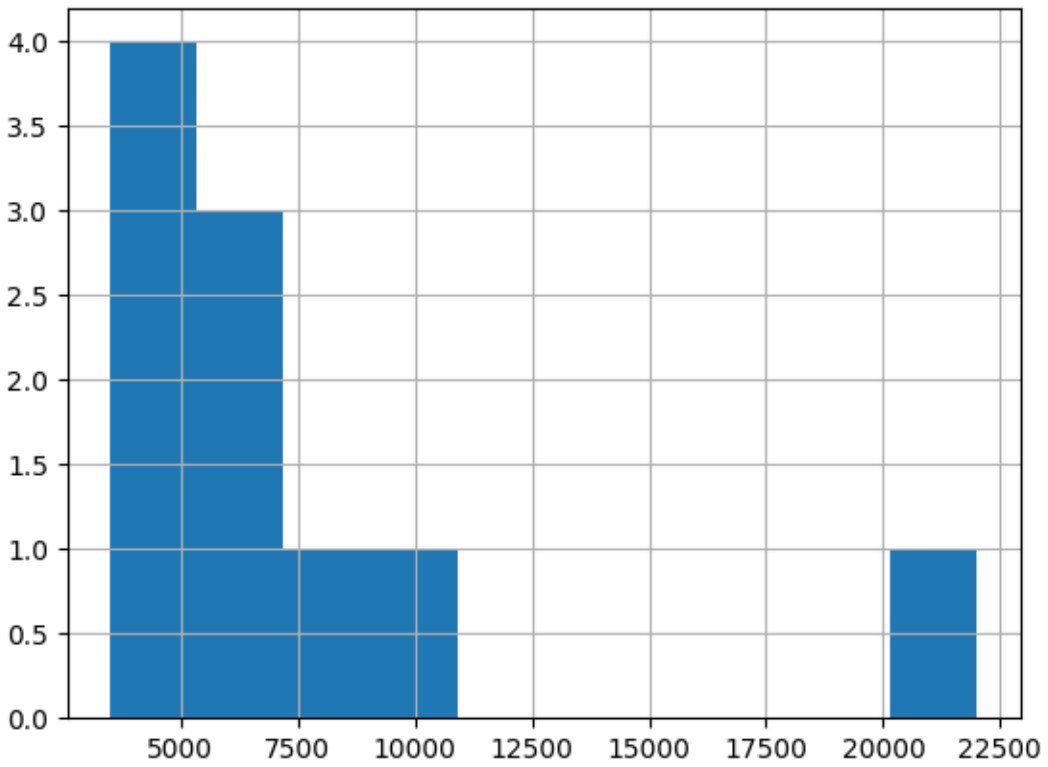
[36]: <Axes: >



[37]:
```
# applying function to a column
new_df["Price"] = new_df["Price"].apply(lambda x : float(x[1:].replace(",",␣
↪"")))
```

[38]:
```
new_df["Price"].hist()
```

[38]: <Axes: >



[39]:
```
new_df.sort_values(by='Price')
```

[39]:

	Make	Colour	Odometer (KM)	Doors	Price
4	Nissan	White	213095	4	3500.0
0	Toyota	White	150043	4	4000.0
5	Toyota	Green	99213	4	4500.0
1	Honda	Red	87899	4	5000.0
8	Toyota	White	60000	4	6250.0
2	Toyota	Blue	32549	3	7000.0
7	Honda	Blue	54738	4	7000.0
6	Honda	Blue	45698	4	7500.0
9	Nissan	White	31600	4	9700.0
3	BMW	Black	11179	5	22000.0

[40]:
```
# Create New Columns
# Series
fuel_type = pd.Series(["Diesel", "Petrol","Diesel", "Petrol","Diesel"])
new_df['Fuel'] = fuel_type
new_df
```

[40]:

	Make	Colour	Odometer (KM)	Doors	Price	Fuel
0	Toyota	White	150043	4	4000.0	Diesel

```
1   Honda    Red          87899      4   5000.0  Petrol
2  Toyota   Blue          32549      3   7000.0  Diesel
3     BMW  Black          11179      5  22000.0  Petrol
4  Nissan  White         213095      4   3500.0  Diesel
5  Toyota  Green          99213      4   4500.0     NaN
6   Honda   Blue          45698      4   7500.0     NaN
7   Honda   Blue          54738      4   7000.0     NaN
8  Toyota  White          60000      4   6250.0     NaN
9  Nissan  White          31600      4   9700.0     NaN
```

[41]:
```
new_df.fillna("Petrol", inplace=True)
```

[42]:
```
new_df
```

[42]:
```
     Make Colour  Odometer (KM)  Doors    Price    Fuel
0  Toyota  White         150043      4   4000.0  Diesel
1   Honda    Red          87899      4   5000.0  Petrol
2  Toyota   Blue          32549      3   7000.0  Diesel
3     BMW  Black          11179      5  22000.0  Petrol
4  Nissan  White         213095      4   3500.0  Diesel
5  Toyota  Green          99213      4   4500.0  Petrol
6   Honda   Blue          45698      4   7500.0  Petrol
7   Honda   Blue          54738      4   7000.0  Petrol
8  Toyota  White          60000      4   6250.0  Petrol
9  Nissan  White          31600      4   9700.0  Petrol
```

```
# python list
fuel_type = [10.0, 6.0, 9.0, 6.0]
new_df['Mileage'] = fuel_type
new_df.head()
```

[43]:
```
# python list
fuel_type = [10.0, 6.0, 9.0, 6.0, 7.3, 9.3, 6.0, 7.3, 9.1, 6.6]
new_df['Mileage'] = fuel_type
new_df.head()
```

[43]:
```
     Make Colour  Odometer (KM)  Doors    Price    Fuel  Mileage
0  Toyota  White         150043      4   4000.0  Diesel     10.0
1   Honda    Red          87899      4   5000.0  Petrol      6.0
2  Toyota   Blue          32549      3   7000.0  Diesel      9.0
3     BMW  Black          11179      5  22000.0  Petrol      6.0
4  Nissan  White         213095      4   3500.0  Diesel      7.3
```

[44]:
```
# Total fuel used
new_df['Total Fuel Used'] = new_df['Odometer (KM)'] / new_df['Mileage']
new_df['Total Fuel Used'] = new_df['Total Fuel Used'].astype(int)
```

```
new_df
```

[44]:

	Make	Colour	Odometer (KM)	Doors	Price	Fuel	Mileage	\
0	Toyota	White	150043	4	4000.0	Diesel	10.0	
1	Honda	Red	87899	4	5000.0	Petrol	6.0	
2	Toyota	Blue	32549	3	7000.0	Diesel	9.0	
3	BMW	Black	11179	5	22000.0	Petrol	6.0	
4	Nissan	White	213095	4	3500.0	Diesel	7.3	
5	Toyota	Green	99213	4	4500.0	Petrol	9.3	
6	Honda	Blue	45698	4	7500.0	Petrol	6.0	
7	Honda	Blue	54738	4	7000.0	Petrol	7.3	
8	Toyota	White	60000	4	6250.0	Petrol	9.1	
9	Nissan	White	31600	4	9700.0	Petrol	6.6	

	Total Fuel Used
0	15004
1	14649
2	3616
3	1863
4	29191
5	10668
6	7616
7	7498
8	6593
9	4787

[45]:
```
# Just Int
new_df['Wheels'] = 4
new_df
```

[45]:

	Make	Colour	Odometer (KM)	Doors	Price	Fuel	Mileage	\
0	Toyota	White	150043	4	4000.0	Diesel	10.0	
1	Honda	Red	87899	4	5000.0	Petrol	6.0	
2	Toyota	Blue	32549	3	7000.0	Diesel	9.0	
3	BMW	Black	11179	5	22000.0	Petrol	6.0	
4	Nissan	White	213095	4	3500.0	Diesel	7.3	
5	Toyota	Green	99213	4	4500.0	Petrol	9.3	
6	Honda	Blue	45698	4	7500.0	Petrol	6.0	
7	Honda	Blue	54738	4	7000.0	Petrol	7.3	
8	Toyota	White	60000	4	6250.0	Petrol	9.1	
9	Nissan	White	31600	4	9700.0	Petrol	6.6	

	Total Fuel Used	Wheels
0	15004	4
1	14649	4
2	3616	4
3	1863	4

```
4            29191       4
5            10668       4
6             7616       4
7             7498       4
8             6593       4
9             4787       4
```

[46]:
```
# remove columns
new_df = new_df.drop("Total Fuel Used", axis=1)
new_df
```

[46]:

	Make	Colour	Odometer (KM)	Doors	Price	Fuel	Mileage	Wheels
0	Toyota	White	150043	4	4000.0	Diesel	10.0	4
1	Honda	Red	87899	4	5000.0	Petrol	6.0	4
2	Toyota	Blue	32549	3	7000.0	Diesel	9.0	4
3	BMW	Black	11179	5	22000.0	Petrol	6.0	4
4	Nissan	White	213095	4	3500.0	Diesel	7.3	4
5	Toyota	Green	99213	4	4500.0	Petrol	9.3	4
6	Honda	Blue	45698	4	7500.0	Petrol	6.0	4
7	Honda	Blue	54738	4	7000.0	Petrol	7.3	4
8	Toyota	White	60000	4	6250.0	Petrol	9.1	4
9	Nissan	White	31600	4	9700.0	Petrol	6.6	4

[47]:
```
# shuffle dataframe
new_df.sample(frac=1)
```

[47]:

	Make	Colour	Odometer (KM)	Doors	Price	Fuel	Mileage	Wheels
9	Nissan	White	31600	4	9700.0	Petrol	6.6	4
1	Honda	Red	87899	4	5000.0	Petrol	6.0	4
2	Toyota	Blue	32549	3	7000.0	Diesel	9.0	4
3	BMW	Black	11179	5	22000.0	Petrol	6.0	4
7	Honda	Blue	54738	4	7000.0	Petrol	7.3	4
4	Nissan	White	213095	4	3500.0	Diesel	7.3	4
5	Toyota	Green	99213	4	4500.0	Petrol	9.3	4
8	Toyota	White	60000	4	6250.0	Petrol	9.1	4
0	Toyota	White	150043	4	4000.0	Diesel	10.0	4
6	Honda	Blue	45698	4	7500.0	Petrol	6.0	4

[48]:
```
new_df.sample(frac=1).reset_index(drop=True)
```

[48]:

	Make	Colour	Odometer (KM)	Doors	Price	Fuel	Mileage	Wheels
0	Honda	Blue	45698	4	7500.0	Petrol	6.0	4
1	Nissan	White	213095	4	3500.0	Diesel	7.3	4
2	Toyota	Green	99213	4	4500.0	Petrol	9.3	4
3	Honda	Blue	54738	4	7000.0	Petrol	7.3	4
4	Nissan	White	31600	4	9700.0	Petrol	6.6	4
5	Toyota	White	60000	4	6250.0	Petrol	9.1	4

6	Toyota	Blue	32549	3	7000.0	Diesel	9.0	4
7	Toyota	White	150043	4	4000.0	Diesel	10.0	4
8	BMW	Black	11179	5	22000.0	Petrol	6.0	4
9	Honda	Red	87899	4	5000.0	Petrol	6.0	4

[49]:
```
# select 20% of data
new_df.sample(frac=0.2)
```

[49]:

	Make	Colour	Odometer (KM)	Doors	Price	Fuel	Mileage	Wheels
2	Toyota	Blue	32549	3	7000.0	Diesel	9.0	4
3	BMW	Black	11179	5	22000.0	Petrol	6.0	4

[ ]:
```

```